BLOCKCHAIN-BASED DIGITAL WARRANTY TRACKER

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING

Use Case Report

submitted by

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CERTIFICATE

This is to certify that the Use Case report entitled **"Blockchain-Based Digital Warranty Tracker"** that is being submitted by **K.M.S.L. Sindhura (23501A0565)**, as part of Assignment-2 for the **Blockchain Technology (23CS4701B)** course in 4-1 during the academic year **2026-27**.

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1. INTRODUCTION

1.1. Overview of Blockchain Technology

Blockchain technology is a decentralized and distributed ledger system that records transactions across a network of independent computers, ensuring transparency, security, and immutability. Unlike a conventional database that is controlled by a single organization, a blockchain operates on a peer-to-peer network in which every participant holds a synchronized copy of the ledger. Each block in the chain stores a cryptographic hash of the block before it, so altering any earlier record would break every block that follows and be detected immediately. Although blockchain was first introduced for cryptocurrency transactions [1], it is now being applied in finance, supply chain management, healthcare, education, and public administration, and it is equally useful for keeping reliable records of consumer products such as warranty information.

1.2. Relevance of Blockchain in Warranty Management

When people purchase electronic products such as mobile phones, laptops, TVs, refrigerators, or headphones, they usually receive a paper or digital invoice and warranty information. The warranty is a promise that the product will be repaired or replaced within a stated period, but proving this later depends on whether the customer can still produce the right document. Customers lose their bills or warranty cards, find it difficult to remember when the warranty expires, and may even be handed fake or modified warranty documents. Service centers, in turn, need a reliable way to verify whether a product is still under warranty, and transferring warranty information to another person when a product is sold is inconvenient.

A Digital Warranty Tracker using blockchain addresses these issues by storing the important warranty information in a tamper-resistant blockchain record. Through a website, the customer can register a purchased product, store its warranty information, check the warranty status, verify whether the warranty record is genuine, and transfer ownership to another person. For example, when Ravi purchases a Dell laptop, the seller enters the Product ID (LAP123456), product name, purchase date (10-09-2026), warranty period (2 years), warranty expiry (10-09-2028), and Ravi's wallet address. The system creates a unique warranty record and stores its important information on the blockchain. [2]

Later, when Ravi's laptop develops a problem, he visits an authorized service center and provides the Product ID. The service center checks the blockchain and sees that the product is genuine, along with its purchase date, warranty expiry, and current owner, and can therefore determine whether the product is covered by warranty. Because the complete invoice and personal information are not stored directly on the blockchain, and only a hash or identifier is kept on-chain while the actual document stays off-chain, the system also keeps cost low and protects privacy. In this way, the blockchain-based warranty tracker links purchase, warranty verification, service, and resale into one verifiable record that customers, sellers, and service centers can trust.

2. BACKGROUND

Traditional warranty handling depends on paper or digital invoices and warranty cards that the customer must keep safe, and on databases that are maintained by sellers or manufacturers. These arrangements lead to problems for customers and service centers alike. Below are some key issues with the existing warranty process:

2.1. Lost Bills and Warranty Cards

Customers usually receive a paper or digital invoice along with warranty information when they buy a product. These bills and warranty cards are often lost, damaged, or deleted. Without them, a customer may find it difficult to prove the purchase and claim the warranty even when the product is still covered.

2.2. Difficulty in Remembering Warranty Expiry

Different products have different warranty periods, such as one year for headphones or two years for a laptop. Customers find it difficult to remember when the warranty of each product expires, and may discover that the warranty has ended only when the product develops a problem.

2.3. Fake or Modified Warranty Documents

Since paper and digital warranty documents can be edited or reproduced, fake or modified warranty documents can be presented, for example with a changed purchase date or an extended expiry date. It is difficult for a service center to judge from the document alone whether it is genuine.

2.4. Verification Burden on Service Centers

Service centers need to verify whether a product is still under warranty before accepting a repair. When the only evidence is a document brought by the customer, the service center must depend on that document or contact the seller, which slows the process and can lead to disagreements.

2.5. Inconvenient Transfer of Warranty on Resale

When a product is sold to another person, transferring the warranty information is inconvenient. The new owner has no easy way to show that the warranty now belongs to them, and there is no record of who owned the product earlier.

2.6. Modification of Records in Normal Databases

A normal database may contain a record such as Product ID: LAP123456, Warranty: 2 years, Expiry: 10-09-2028. If someone with sufficient database privileges changes this record, it may be difficult for an ordinary customer to detect the change. There is also no built-in auditable history showing what was changed and when.

Due to these challenges, there is a strong need for a secure, verifiable, and automated warranty management system. Blockchain technology offers a promising solution by providing

tamper-resistant records, easy verification, and transfer of ownership with a maintained transaction history.

3. BLOCKCHAIN BASICS

Blockchain is a distributed ledger technology that enables secure, transparent, and tamper-resistant recording of transactions without relying on a single trusted authority. Instead of storing data in one database, a blockchain replicates it across many nodes that follow common rules to agree on the state of the ledger. This makes it well suited to a warranty tracker, where the main requirement is tamper resistance and verification. Below are the key concepts that define blockchain technology:

3.1. Decentralization

A traditional warranty database is controlled by one organization, so a person with sufficient privileges can change a record. A blockchain, on the other hand, runs on a decentralized network where multiple independent nodes store and validate the data. No single participant has full control over the warranty records, and the data can be checked independently by customers and service centers. [1]

3.2. Immutability

Once a transaction is recorded on a blockchain and confirmed by the network, it cannot easily be altered or deleted. Each block contains the hash of the previous block, forming a chain, so changing one record would require changing every later block. For warranties, this means the purchase date, expiry date, and ownership of a product cannot be modified without the change being noticed, which makes unauthorized modification harder.

3.3. Transparency and Auditability

Every authorized participant can read the same ledger, and every transaction can be audited. When a warranty is registered or ownership is transferred, the transaction is recorded on the blockchain, creating an auditable record that customers, sellers, and service centers can all check.

3.4. Smart Contracts

Smart contracts are self-executing programs stored on the blockchain that run automatically when predefined conditions are met. [2] In the warranty tracker, a smart contract written in Solidity creates the warranty record at registration, returns the stored details when a Product ID is verified, determines whether the warranty is ACTIVE or EXPIRED by comparing the expiry date with the current date, and changes the owner when ownership is transferred. The contract needs only a few functions: registerWarranty(), verifyWarranty(), getWarranty(), and transferOwnership().

3.5. Consensus Mechanisms

Consensus mechanisms are the rules by which the nodes of a blockchain agree on which transactions are valid and in which order they are added. Public networks such as Ethereum rely on such mechanisms to keep all copies of the ledger identical. For the prototype in this report, a local blockchain (Ganache) or an Ethereum test network is used, so that the working of the system can be demonstrated without real cost.

3.6. Cryptographic Security

Blockchain security is based on public-key cryptography and hash functions. Every user has a wallet with a private key used to sign transactions and a public address that identifies the account, so only the genuine owner of a wallet can approve an action such as transferring a warranty. A hash function converts any document into a fixed-length digital fingerprint, so even a small change in an invoice produces a completely different hash. For privacy and lower cost, the complete invoice or personal information is not stored directly on a public blockchain; only a hash or identifier is stored on-chain while the actual document is kept off-chain. [1]

3.7. Public Blockchains, Fees and Wallets

On a public blockchain such as Ethereum, anyone can read the ledger and transactions can have fees. Users need a wallet, such as MetaMask, to interact with the blockchain and to sign transactions. [4] These features make a public network convenient for demonstration but mean that sensitive data should never be written to it directly. The prototype in this report is developed using Solidity, Ganache, MetaMask, and ethers.js.

By combining these features (decentralization, immutability, transparency, smart contracts, and cryptographic security), blockchain provides a sound foundation for warranty management. It ensures that warranty records are tamper-resistant, verifiable, and updated through transparent and automated processes.

4. USE CASE OVERVIEW

The use case for a Blockchain-Based Digital Warranty Tracker aims to improve the way product warranties are recorded and checked by using blockchain technology. The system provides a tamper-resistant and verifiable way to register a product warranty, check its status, verify that the record is genuine, and transfer ownership to another person.

4.1. Objectives

The primary objective of this system is to reduce the dependency on physical warranty cards by storing important warranty information in a tamper-resistant blockchain record. Through a website, a customer can register a purchased product, store its warranty information, check the warranty status, verify whether the warranty record is genuine, and transfer ownership to another person. Each product is represented by a warranty record containing the Product ID, product name, purchase date, expiry date, and current owner.

A second objective is to make warranty verification easy for service centers. A service center only needs to enter the Product ID to see whether the record is genuine, the purchase date, the warranty expiry, the current owner, and whether the warranty is ACTIVE or EXPIRED, which reduces disputes between customers and service centers. In the example of Ravi's Dell laptop (Product ID LAP123456, purchased on 10-09-2026 with a 2-year warranty expiring on 10-09-2028), the service center can confirm the warranty directly from the blockchain.

A third objective is to allow ownership transfer. If Ravi sells his laptop to Priya, the ownership of the warranty record moves from Ravi to Priya, and the blockchain maintains the transaction history. The system also aims to provide an auditable history and to make unauthorized modification of warranty records harder, while keeping personal information and complete invoices off-chain for privacy and lower cost.

4.2. Scope of the System

The blockchain-based warranty system focuses on registering, verifying, and transferring product warranties. The system includes:

1. **Manufacturer / Seller:** Registers products, creates warranty records, and activates the warranty for the customer.
2. **Customer:** Views purchased products, checks the warranty expiry, views warranty details, and transfers ownership to another person.
3. **Service Center:** Searches by Product ID, verifies the warranty, and checks the current owner and status.
4. **Blockchain Network:** Stores the warranty records in a tamper-resistant way and executes the smart contract rules.

4.3. System Architecture

The architecture of the Blockchain-Based Digital Warranty Tracker is illustrated in Figure 4.1 and consists of the following components:

4.3.1. User Layer

This layer consists of the primary users who interact with the system. (Figure 4.1 - User Layer)

- **Manufacturer / Seller:**
  - Register products and create warranty records.
  - Activate the warranty against the customer's wallet address.
- **Customer:**
  - View purchased products and warranty details.
  - Check the warranty expiry and transfer ownership.
- **Service Center:**
  - Search the Product ID.
  - Verify the warranty and check the current owner and status.

4.3.2. Web Application Layer

This layer acts as the interface between users and the blockchain. (Figure 4.1 - Web Application Layer)

- **Web Application:**
  - Allows the seller to register warranties and the customer or service center to verify them.
  - Allows customers to view their products and transfer ownership.
- **Backend / Database (optional):**
  - Stores additional information such as invoices off-chain, using Node.js with Express and MongoDB or Firebase.
  - For a very minimal version, the backend can be removed and the web page can talk to the blockchain directly.

4.3.3. Blockchain Layer

This layer ensures tamper resistance and verification of warranty records. (Figure 4.1 - Blockchain Layer)

- **Smart Contract:**
  - Stores the warranty details: Product ID, Warranty ID, purchase date, expiry date, and owner.
  - Manages registration, verification, status calculation, and ownership transfer.
- **Blockchain Network:**
  - Maintains the tamper-resistant record of all warranties and their owners.
  - Keeps the transaction history of ownership transfers.

- Only a hash or identifier is stored on-chain; the actual invoice and personal information stay off-chain.

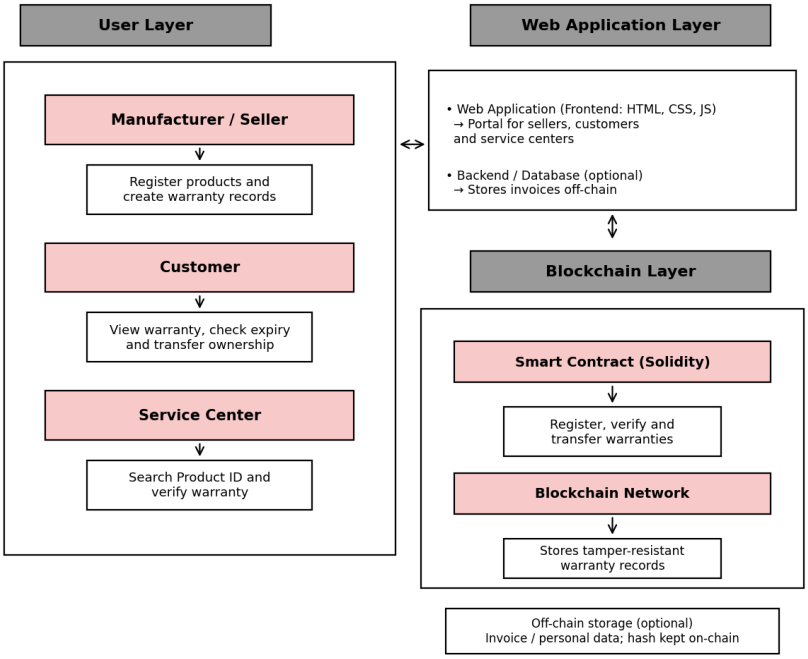


Figure 4.1: Architecture of Blockchain-Based Digital Warranty Tracker

5. IMPLEMENTATION

5.1. Setting Up the Blockchain Environment

To implement the blockchain-based warranty tracker, Ethereum is used as the blockchain platform. The smart contract is written in Solidity (v0.8.19), whose built-in overflow checks add to the safety of the code. [3] Ganache provides a personal Ethereum blockchain for local testing, and an Ethereum test network can be used for further testing. MetaMask is used as the wallet to manage accounts and sign transactions, and ethers.js is used by the frontend to communicate with the contract. The frontend is built using HTML, CSS, and JavaScript. A Node.js and Express backend with MongoDB or Firebase is optional. For a very minimal version, only HTML, JavaScript, Solidity, Ganache, and MetaMask are needed.

5.2. Defining Smart Contracts

5.2.1. Writing the Smart Contract for Warranty Records

The WarrantyTracker contract is responsible for registering warranties, verifying them, and transferring ownership. It defines a Warranty struct that holds the product ID, product name, current owner, purchase date, expiry date, and a hash of the off-chain invoice. Mappings store the warranty of each product, the authorized sellers, and the ownership history of each product, so that every warranty stays linked to its current owner and the history of transfers can be traced.

```
struct Warranty {
    string  productId;
    string  productName;
    address owner;
    uint256 purchaseDate;
    uint256 expiryDate;
    string  documentHash;
    bool    exists;
}

address public admin;
mapping(address => bool)      internal sellers;
mapping(bytes32 => Warranty)  internal warranties;
mapping(bytes32 => address[]) internal ownershipHistory;
```

5.2.2. Product Registration (Module 1)

Only an authorized manufacturer or seller can create warranty records. The account that deploys the contract becomes the administrator and the first seller, and can add other sellers. The seller registers a product by calling registerWarranty with the Product ID, product name, customer wallet address, purchase date, expiry date, and the hash of the invoice. The function rejects a Product ID that has already been registered, creates the warranty record, and stores the customer as the first owner. For the example in this report, the seller enters LAP123456, Dell Laptop, 10-09-2026, a warranty of 2 years (expiry 10-09-2028), and Ravi's wallet address.

```
modifier onlySeller() {
    require(sellers[msg.sender], "Not an authorised seller");
    _;
}

constructor() {
```

```
        admin = msg.sender;
        sellers[msg.sender] = true;
    }

    function addSeller(address seller) external {
        require(msg.sender == admin, "Only admin");
        sellers[seller] = true;
    }

    function registerWarranty(
        string memory productId, string memory productName,
        address customer, uint256 purchaseDate,
        uint256 expiryDate, string memory documentHash
    ) external onlySeller {
        bytes32 key = keccak256(bytes(productId));
        require(!warranties[key].exists, "Product already registered");
        require(expiryDate > purchaseDate, "Invalid warranty period");
        warranties[key] = Warranty(productId, productName, customer,
            purchaseDate, expiryDate, documentHash, true);
        ownershipHistory[key].push(customer);
    }
```

5.2.3. Warranty Verification and Status (Modules 2 and 3)

The customer or service center enters the Product ID, and the getWarranty function returns the stored product details. The verifyWarranty function determines the status automatically: the warranty is ACTIVE if the warranty expiry date is later than the current date, and EXPIRED if the expiry date has passed. If the Product ID is not found, no genuine record exists. The getOwnershipHistory function returns every owner of the product. These are read-only functions and can be called freely.

```
    function getWarranty(string memory productId)
        external view returns (Warranty memory)
    {
        bytes32 key = keccak256(bytes(productId));
        require(warranties[key].exists, "Warranty not found");
        return warranties[key];
    }

    function verifyWarranty(string memory productId)
        external view returns (string memory)
    {
        bytes32 key = keccak256(bytes(productId));
        require(warranties[key].exists, "Warranty not found");
        return block.timestamp < warranties[key].expiryDate
            ? "ACTIVE" : "EXPIRED";
    }

    function getOwnershipHistory(string memory productId)
        external view returns (address[] memory)
    {
        return ownershipHistory[keccak256(bytes(productId))];
    }
```

5.2.4. Ownership Transfer (Module 4)

If Ravi sells the laptop to Priya, Ravi calls transferOwnership with the Product ID and Priya's wallet address. The function checks that the caller is the current owner, changes the owner to Priya, and adds her to the ownership history. The earlier owner remains visible in the history, so the blockchain maintains the transaction history.

```
function transferOwnership(string memory productId, address newOwner)
    external
{
    bytes32 key = keccak256(bytes(productId));
    Warranty storage w = warranties[key];
    require(w.exists, "Warranty not found");
    require(w.owner == msg.sender, "Not the owner");
    require(newOwner != address(0), "Invalid new owner");
    w.owner = newOwner;
    ownershipHistory[key].push(newOwner);
}
```

5.3. Deploying and Integrating with a Frontend

After developing the smart contract, it is deployed to Ganache (or an Ethereum test network) and connected to the web page using ethers.js and MetaMask. [4] The frontend contains a Seller Dashboard for registering warranties and a Verification Page for checking them. The expected output of the two pages is shown below.

```
----------------------------------
       REGISTER WARRANTY
----------------------------------
Product ID:       LAP123456
Product Name:     Dell Laptop
Purchase Date:    10-09-2026
Warranty:         2 Years

        [ REGISTER ]
----------------------------------
```

Seller Dashboard (Register Warranty)

```
----------------------------------
       WARRANTY VERIFICATION
----------------------------------
Enter Product ID:

LAP123456

        [ VERIFY ]
----------------------------------
Product: Dell Laptop
Status: ACTIVE
Purchase Date: 10-09-2026
Expiry Date: 10-09-2028
----------------------------------
```

Verification Page

5.4. Workflow

The workflow of the Blockchain-Based Digital Warranty Tracker, as illustrated in Figure 5.1, details the interactions between the seller, customer, and service center to ensure secure registration, verification, and transfer of warranties.

The process begins with the seller registering the product and its warranty against the customer's wallet address. The customer can then view the product and its warranty details at any time and, when the product is sold, transfer ownership to the new owner. When the product develops a problem, the customer approaches an authorized service center and provides the Product ID. [2]

The service center enters the Product ID, and the smart contract checks the blockchain record. As per Figure 5.1, if the record is found and the expiry date is later than the current date, the warranty is shown as ACTIVE together with the purchase date, expiry date, and current owner. If the expiry date has passed, the warranty is shown as EXPIRED, and if no record exists, the warranty cannot be verified as genuine.

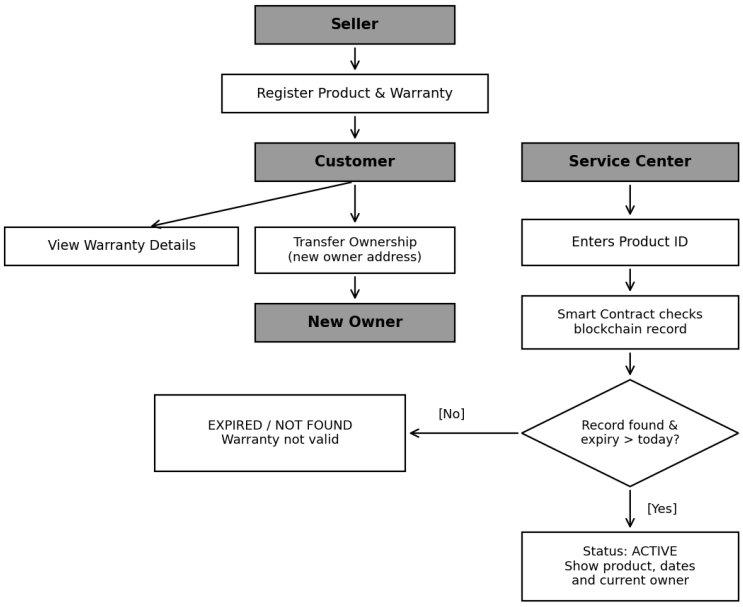


Figure 5.1: Workflow of Blockchain-Based Digital Warranty Tracker

6. BENEFITS

Blockchain technology enhances warranty management by introducing tamper resistance, easy verification, and automation. Traditional warranty handling relies on bills and warranty cards that can be lost or altered. By using blockchain, the warranty process becomes easier to verify and easier to trust. Below are the key advantages:

6.1. Reduced Dependency on Physical Warranty Cards

Because the important warranty information is stored in a blockchain record, customers no longer depend only on paper bills or warranty cards. The warranty can be checked by entering the Product ID.

6.2. Easier Warranty Verification

A service center can verify whether a product is genuine and under warranty directly from the blockchain by entering the Product ID. The status is calculated automatically as ACTIVE or EXPIRED by comparing the expiry date with the current date.

6.3. Auditable History

Since warranty registration and ownership transfers are recorded through a smart contract, the blockchain provides an auditable history showing who owned the product and when ownership changed.

6.4. Unauthorized Modification Made Harder

In a normal database, a person with sufficient privileges may change a record without an ordinary customer being able to detect it. With blockchain, the stored record cannot be silently changed, which makes unauthorized modification of purchase and expiry dates harder.

6.5. Ownership Transfer

When a product is sold, the warranty can be transferred to the new owner, such as from Ravi to Priya, and the blockchain maintains the transaction history. This removes the inconvenience of passing on warranty papers.

6.6. Fewer Disputes Between Customers and Service Centers

When the customer and the service center check the same tamper-resistant record, there is less scope for argument over the purchase date, the expiry date, or the current owner, which reduces disputes.

Implementing blockchain in warranty management makes the verification of warranty simple and reliable. It reduces dependence on physical documents, supports ownership transfer, and gives customers and service centers a common, auditable record.

7. CHALLENGES

While a blockchain-based warranty tracker offers several advantages, it also has limitations that must be stated honestly and considered for effective implementation. Below are the key challenges:

7.1. Accuracy of the Information Entered Initially

Blockchain cannot prove that the information entered initially is truthful. If the seller enters a wrong purchase date or warranty period, the blockchain will preserve that entry as it is. The correctness of the data therefore still depends on the authorized seller or manufacturer who registers the warranty.

7.2. Transaction Fees

Public blockchain transactions can have fees. Registering a warranty or transferring ownership on a public network therefore involves a cost, which must be considered when choosing the network for real use. Testing on Ganache or a test network avoids this cost.

7.3. Wallet Requirement

Users need a wallet, such as MetaMask, for blockchain interaction. Customers who are not familiar with wallets may find it difficult to register, view, or transfer warranties without guidance.

7.4. Privacy of Personal Information

Personal information should not be stored directly on-chain, because a public blockchain can be read by anyone. The complete invoice and customer details must be kept off-chain, and only a hash or identifier should be stored on the blockchain.

7.5. Integration with Manufacturers

Real manufacturers would need to integrate their systems with the platform before it can be used on a large scale. Until then, the system works only for the sellers and service centers that participate in it.

Despite these challenges, blockchain-based warranty tracking offers a practical way to make warranty records verifiable. To overcome these limitations, attention must be given to the accuracy of the entered data, protection of personal information, and integration with manufacturers.

8. CONCLUSION

The report explores the use of blockchain technology for warranty management, addressing key problems of the traditional process. It highlights the problems of lost bills and warranty cards, difficulty in remembering warranty expiry, fake or modified warranty documents, the effort needed by service centers to verify warranties, and the inconvenience of transferring warranty information when a product is sold. By storing important warranty information in a tamper-resistant blockchain record, the proposed system makes warranty verification easier and more reliable.

The implementation, as described in the report, provides a simple structure with four modules: product registration, warranty verification, warranty status, and ownership transfer. Sellers register products, customers and service centers verify the warranty using the Product ID, the status is calculated automatically as ACTIVE or EXPIRED, and ownership can be transferred, for example from Ravi to Priya, with the transaction history maintained. The system reduces dependency on physical warranty cards, provides an auditable history, makes unauthorized modification harder, and reduces disputes between customers and service centers.

8.1. Future Outlook for Enhancements

To further improve the system, a QR code can be placed on the product or invoice so that the warranty can be checked by scanning it. A mobile application can be developed for customers, and automatic warranty-expiry notifications can remind them before the warranty ends. Integration with manufacturer APIs can allow warranty records to be created automatically when a product is sold, and AI-based invoice extraction can read the details from an invoice and reduce manual entry. IoT-based product registration can register a product automatically, and multi-manufacturer support can allow products of different brands to be tracked on one platform. By adding these enhancements, the Blockchain-Based Digital Warranty Tracker can grow into a complete solution for warranty management.

9. SDG's ADDRESSED

The Blockchain-Based Digital Warranty Tracker aligns with several United Nations Sustainable Development Goals (SDGs) by supporting reliable service for consumers, encouraging digital innovation, and promoting transparency. Below are the key SDGs addressed and their justifications:

SDG 12: Responsible Consumption and Production

A reliable and easily verified warranty makes it simpler for customers to get their products repaired under warranty and to resell them with a transferable warranty. This encourages longer use of electronic products and reduces the use of paper bills and warranty cards.

SDG 9: Industry, Innovation, and Infrastructure

The use of blockchain and smart contracts for warranty registration and verification promotes digital innovation in after-sales services. Automating warranty checks encourages the adoption of modern and secure digital technologies by sellers, manufacturers, and service centers.

SDG 16: Peace, Justice, and Strong Institutions

Tamper-resistant and auditable warranty records reduce fake or modified warranty documents and lower disputes between customers and service centers. Verifiable records help settle disagreements fairly and build trust in the business process.

By integrating blockchain, the Digital Warranty Tracker aligns with these SDGs, fostering a transparent and trustworthy system for managing product warranties.

10. REFERENCES

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[4] MetaMask Documentation. Visited on October 25, 2026, from https://docs.metamask.io

[5] Ethers.js Documentation. Visited on October 25, 2026, from https://docs.ethers.org

11. APPENDIX A

Video Presentation:
https://drive.google.com/file/d/1c4y-iwBNYYNdogS_fvV_RpEI8mKwlSuR/view?usp=sharing

Github Repository:
https://github.com/Sindhura-Karumuri/BlockChain-based-Digital-Warranty-Tracker.git